

1 P3 C2 — Triangular maximum depths

Let B be the Bulgarian-solitaire move on partitions. Write $T_k = k(k+1)/2$, and let $D_B(n)$ be the maximum, over partitions of n , of the number of moves before first reaching a cyclic partition. For every $k \geq 1$,

$$D_B(T_k) = k(k-1).$$

The upper proof covers every starting partition; the lower proof exhibits a partition attaining the bound. At triangular size the only cyclic partition is the fixed staircase $(k, k-1, \dots, 1)$. The general upper and lower bounds below are written mathematical proofs. The associated Lean development establishes triangular convergence and uniqueness for all k , exact maximum depths for $k = 1, \dots, 6$, and the abstract arithmetic used in the upper proof; the pile-lifetime application has not been formalized in Lean.

1.1 1. Sharp upper bound

For every $k \geq 1$ and every partition λ of $T_k = k(k+1)/2$, Bulgarian solitaire reaches the staircase $(k, k-1, \dots, 1)$ in at most $k(k-1)$ moves. Together with the explicit [lower-bound witness](#), this proves $D_B(T_k) = k(k-1)$ as a **written mathematical proof**. The general upper bound in this document has not been formalized in Lean; the Lean package establishes the triangular destination and finite cases.

The argument reconstructs the upper-bound half of [Griggs and Ho, Theorem 3.7](#). It proves the needed timing facts directly using pile lifetimes. These correspond to their Proposition 3.2(1),(3) and Lemmas 3.3–3.6; their Proposition 3.2(2) is not needed for this route. We do not use Theorem 3.7 as a premise.

1.1.1 Pile lifetimes and a sequence bound

Write c_i for the number of piles in $B^{i-1}(\lambda)$, with $i \geq 1$. An original pile of size a exists at times $1, \dots, a$. The pile created by move i has size c_i and exists at times

$$J_i = [i+1, i+c_i].$$

Its last time is $e_i = i+c_i$. At any time m , precisely c_m of these original-pile and created-pile intervals contain m . One new interval starts between times i and $i+1$; if d_i intervals end at i , then

$$c_{i+1} = c_i + 1 - d_i.$$

(1)

In particular $c_{i+1} \leq c_i + 1$; a rise by one means **no** pile dies, and a constant pair means **exactly one** pile dies.

We use a *sandwich pattern* of level x and width $L = q - p \geq 2$:

$$(c_p, c_{p+1}, \dots, c_q) = (x-1, x, \dots, x, x+1).$$

(2)

The two rises occur at its ends. The following elementary sandwich rule will locate one. If $i < j$, $c_i \leq x - 1$, and $c_j \geq x + 1$, choose p as the last index before j with $c_p \leq x - 1$, and q as the first subsequent index with $c_q \geq x + 1$. Equation (1) forces $c_p = x - 1$, $c_{p+1} = \dots = c_{q-1} = x$, and $c_q = x + 1$. Thus (2) occurs within $[i, j]$.

1.1.2 The retreat lemma

Suppose (2) occurs and $p > x$. Among the x recent created piles $J_{p-x}, \dots, J_{p-1}$, at least one is absent at time p , since $c_p = x - 1$. The last, J_{p-1} , is present. Let $u \in [p - x, p - 2]$ be the largest index whose interval is absent. All of $J_{u+1}, \dots, J_{p-1}$ are present at p . Since $c_{p+1} = c_p + 1$, none dies at p , so they remain present at $p + 1$.

Put $h = p - u - 1$ and $y = h + 1 = p - u \leq x$. The absence of J_u at p gives $c_u \leq h$. The presence of J_{u+1} at $p + 1$ gives $c_{u+1} \geq h + 1$. Applying $c_{u+1} \leq c_u + 1$ shows

$$c_u = h = y - 1, \quad c_{u+1} = h + 1 = y.$$

(3)

Moreover J_{u+1} ends at time $u + 1 + y = p + 1$.

If $L = 2$, the pattern also has $c_{p+2} = c_{p+1} + 1$, so no interval ends at $p + 1$, a contradiction. Hence every width-two pattern satisfies

$$p \leq x.$$

(4)

Now let $L \geq 3$. As long as no later value c_{u+s} has risen from y to $y + 1$, the consecutive intervals

$$J_{u+s}, J_{u+s+1}, \dots, J_{p+s-1}$$

are all present at time $p + s$, for $1 \leq s \leq L - 1$. This holds for $s = 1$ by the choice of u and the lack of a death at p . If $c_{u+s} = y$, then J_{u+s} ends at $(u + s) + y = p + s$. For $s \leq L - 2$, the pair $c_{p+s} = c_{p+s+1} = x$ permits exactly one death. It must be this interval; all the others survive, and J_{p+s} is born. This proves the induction. At each stage the active J_{u+s} implies $c_{u+s} \geq y$, while (1) and the preceding value y imply $c_{u+s} \leq y + 1$. Hence it either stays at y or supplies the desired rise.

At $s = L - 1$, the transition from $c_{q-1} = x$ to $c_q = x + 1$ permits no death. Thus J_{u+L-1} cannot have size y , which would make it end at $q - 1$. A first rise occurs at some v with

$$u + 2 \leq v \leq u + L - 1, \quad (c_u, \dots, c_v) = (y - 1, y, \dots, y, y + 1).$$

(5)

It occurs by $p+1$: otherwise both c_p and c_{p+1} would equal y , contradicting their values $x-1, x$. Thus $u \geq p-x$, $v \leq p+1$, $y \leq x$, and the new width $v-u < L$. This is the retreat step.

Each retreat decreases the positive integer width. It therefore stops either at a pattern with start at most its level or at width two, where (4) gives the same bound. Working backward through $p \leq u+x$, while levels never increase, gives the explicit bound

$$\boxed{p \leq x(L-1)}$$

(6)

for every sandwich pattern (2). More formally, if $p \leq x$, then $p \leq x(L-1)$ since $L \geq 2$. Otherwise induction on L gives $p \leq u+x \leq y(L'-1) + x \leq x(L-2) + x = x(L-1)$ for the smaller width $L' < L$. This abstract induction is also kernel-checked as [sandwich_start_bound](#); the lifetime argument above proves its application hypotheses in writing.

1.1.3 A special full-width pattern

We also need one bound that uses the number n of cards. Suppose

$$(c_p, \dots, c_{p+k}) = (k-2, k-1, \dots, k-1, k).$$

(7)

Here $k \geq 3$. The interval J_p ends at $p+k-2$. Since $c_{p+k-2} = c_{p+k-1} = k-1$, it is the **only** interval ending then. No interval ends at $p+k-1$, since the next count rises from $k-1$ to k .

At time $p+k$, the $k-1$ intervals $J_{p+1}, \dots, J_{p+k-1}$ are all present. There is exactly one other pile. If it is original, its original size is at least $p+k$, and hence $n \geq p+k$. If it is J_1 , then $1+c_1 \geq p+k$, and $c_1 \leq n$ gives $n+1 \geq p+k$.

The only other possibility would be J_i for some $2 \leq i \leq p-1$; J_p has already died. Such a J_i is also the unique extra pile at time $p+k-1$, alongside $J_{p+1}, \dots, J_{p+k-2}$. Hence J_{i-1} is absent at that time, so $(i-1) + c_{i-1} \leq p+k-2$. But J_i survives to $p+k$, and (1) yields

$$(i-1) + c_{i-1} \geq (i-1) + (c_i - 1) \geq p+k-2.$$

Equality follows, making J_{i-1} a second interval ending at $p+k-2$, contrary to the uniqueness of J_p . We have proved

$$\boxed{p+k \leq n+1}.$$

(8)

1.1.4 Finding the final pattern

Every trajectory at triangular size eventually reaches the staircase, by [triangular convergence](#). Let t be the **first** move at which it does, so $B^t(\lambda)$ is the staircase. If $t = 0$, there is nothing to prove. Otherwise $c_{t+1} = k$. The fresh pile made on move t has size c_t , so $c_t \leq k$. If $c_t = k$,

then its predecessor has k piles and must itself be the staircase: after removing one from each, its surviving piles must be exactly $k-1, k-2, \dots, 1$, with one exhausted pile. This contradicts minimality of t . Equation (1) now forces $c_t = k-1$, and the rise to $c_{t+1} = k$ entails no death at time t .

Assume $t \geq k+1$. Of the k intervals $J_{t-k}, \dots, J_{t-1}$, at least one is absent at time t , because only $k-1$ piles exist. The last, J_{t-1} , is present. Choose the largest absent index $u \in [t-k, t-2]$. Repeating the argument leading to (3), now with $p = t$ and $x = k$, gives

$$c_u = t - u - 1, \quad c_{u+1} = t - u.$$

(9)

If $u \geq t - k + 1$, then $c_u \leq k - 2$. Apply the sandwich rule between u and $t + 1$ at level $k - 1$. It produces a type-II pattern $k - 2, k - 1, \dots, k - 1, k$ with endpoints $p \geq t - k + 1$, $q \leq t + 1$, and width $q - p \leq k$. When that width is exactly k , it is the special pattern (7).

The remaining case is $u = t - k$, giving $c_{t-k} = k - 1$ and $c_{t-k+1} = k$. If some $c_j \leq k - 2$ for $t - k + 2 \leq j \leq t - 1$, the same sandwich rule again produces a type-II pattern within $[t - k + 1, t + 1]$. If instead some such $c_j \geq k + 1$, the rule at level k , between $t - k$ and j , produces a pattern

$$(c_p, \dots, c_q) = (k - 1, k, \dots, k, k + 1), \quad t - k \leq p < q \leq t - 1, \quad q - p \leq k - 1.$$

(10)

Finally suppose every intermediate c_j is $k - 1$ or k . By the choice $u = t - k$, all $J_{t-k+1}, \dots, J_{t-1}$ are present at time t . They account for all $k - 1$ piles, and all survive to $t + 1$; the only new pile is J_t , of size $k - 1$. At time $t + 1$, each older J_j has remaining size $j + c_j - t \leq (t - 1) + k - t = k - 1$. No pile can have size k , contradicting the staircase. Thus either (10) occurs, or a type-II pattern occurs with width at most k .

1.1.5 The deadline

For $k = 1$, the only partition is fixed. For $k = 2$, the three partitions of 3 have depths 1, 0, 2, respectively: $(3) \rightarrow (2, 1)$, $(2, 1)$ is fixed, and $(1, 1, 1) \rightarrow (3) \rightarrow (2, 1)$. Now let $k \geq 3$. If $t \leq k$, then $t \leq k(k - 1)$.

Suppose $t \geq k + 1$. If a type-II pattern has full width k , it is (7), and its allowed index range forces $p = t - k + 1$, $q = t + 1$. By (8), $t = p + k - 1 \leq n = T_k \leq k(k - 1)$, the last inequality holding for $k \geq 3$.

For every other pattern just found, its level x is k or $k - 1$, its width L is at most $k - 1$, and its start satisfies $p \geq t - k$. Equation (6) gives

$$t \leq p + k \leq x(L - 1) + k \leq k(k - 2) + k = k(k - 1).$$

The lower-bound witness attains this deadline, so the maximum first cyclic depth at T_k is exactly $k(k - 1)$.

1.2 2. Attaining construction

Let $k \geq 2$, $n = T_k = k(k+1)/2$, and B be the Bulgarian-solitaire move. The partition

$$\lambda^{(0)} = (k-1, k-1, k-2, \dots, 2, 1, 1)$$

has first cyclic depth exactly $k(k-1)$. The notation includes three copies of 1 when $k = 2$, so $\lambda^{(0)} = (1, 1, 1)$ in that case. Consequently the maximum depth satisfies $D_B(T_k) \geq k(k-1)$. For $k = 1$, the sole partition (1) is fixed and gives the same lower bound 0. This proves the lower-bound half only; it does not establish an upper bound for arbitrary starting partitions.

Use zero-based columns j and one-based rows i . A cell has diagonal index $i+j$. Let S_k be the staircase diagram: every cell on diagonals $1, \dots, k$ is occupied, so column $j < k$ has height $k-j$. For $0 \leq p < k$ and $0 \leq q \leq k$, form $D(p, q)$ by deleting the diagonal- k cell in column p and adding the diagonal- $(k+1)$ cell in column q . It has T_k cells. The added cell leaves a vertical gap when $q = p$. Otherwise its column heights, including a possible zero at the end, are

$$h_j = k - j - \mathbf{1}_{j=p} + \mathbf{1}_{j=q} \quad (0 \leq j \leq k).$$

These heights are weakly decreasing exactly when $q \neq p+1$. Indeed the only way the deletion and addition can reverse an adjacent pair is to decrease column p and increase column $p+1$. Thus $D(p, q)$ is a Ferrers partition whenever $q \notin \{p, p+1\}$. In particular, $D(0, k) = \lambda^{(0)}$.

Before sorting, a Bulgarian move maps a card (i, j) with $i > 1$ to $(i-1, j+1)$, and a top card $(1, j)$ to $(j+1, 0)$. Hence on diagonal w it rotates column j to $j+1 \pmod w$. Every full diagonal of $D(p, q)$ remains full. The hole and added card move respectively to columns $p+1 \pmod k$ and $q+1 \pmod{k+1}$. If the resulting D is Ferrers, it is already in decreasing pile order, so the sorting part of B changes nothing.

Put $T = k(k-1)$ and, for $0 \leq t < T$, set

$$p_t = t \bmod k, \quad q_t = (k+t) \bmod (k+1).$$

Write $t = ak + b$, where $0 \leq a \leq k-2$ and $0 \leq b < k$. Then $p_t = b$ and, because $k \equiv -1 \pmod{k+1}$,

$$q_t = \begin{cases} b - a - 1, & b \geq a + 1, \\ k + b - a, & b \leq a. \end{cases}$$

In the first case $q_t < b$; in the second $q_t - b = k - a \geq 2$. Therefore $q_t \neq p_t$ and $q_t \neq p_t + 1$ throughout $0 \leq t < T$. Each $D(p_t, q_t)$ is thus a partition. Starting with $D(p_0, q_0) = \lambda^{(0)}$, the diagonal rotation and the Ferrers criterion give, by induction,

$$B^t(\lambda^{(0)}) = D(p_t, q_t) \quad (0 \leq t < T).$$

At $t = T - 1 = (k-2)k + (k-1)$, we have $p_t = k-1$ and $q_t = 0$, so the partition is $(k+1, k-1, k-2, \dots, 2)$, with the descending tail empty for $k = 2$. Its next *unsorted* move is $(k-1, k, k-2, \dots, 1)$: this is $D(0, 1)$, since $p_T = 0$ and $q_T = 1$. Sorting its first two entries yields the staircase $(k, k-1, \dots, 1)$, which is fixed by B . This also shows explicitly that the first sorting change occurs on the move numbered T .

For every $t < T$, $D(p_t, q_t)$ still has an occupied cell on diagonal $k + 1$, whereas the staircase has none. The staircase is the **only cyclic partition** of T_k , by the general triangular uniqueness theorem proved in [TriangularGeneral.lean](#) and explained in the [C1 proof](#). Thus no earlier state is cyclic, and the first cyclic depth of this witness is exactly $T = k(k - 1)$.

The witness and lower-bound result appear in [Griggs and Ho, Theorem 3.1](#). The modular calculation above supplies an explicit check that the unsorted diagonal rotations remain valid partitions at every earlier step.

1.3 Verification and provenance

The general triangular destination and cyclic uniqueness are proved in [TriangularGeneral.lean](#). The exact finite maximum-depth theorems are in the [P3 Lean package](#), and the abstract sandwich induction is in [C2SandwichBound.lean](#). The [recorded Lean replay](#) is finite or abstract evidence as stated, not a Lean formalization of the all- k depth result.

The sharp result is classical. The proof cites Griggs and Ho’s original preprint for historical context and reconstructs its needed upper-bound steps rather than assuming the target theorem.